PLECS Setup

This document walks you through installing PLECS and everything needed to run your controller on the LaunchPad: simulate in PLECS, generate C code automatically, flash it onto the F28P55x over USB, and then watch signals and tune parameters live while it runs. Plexim's own documentation is good - this guide tells you what to install, in what order, with what settings, and points you to the right manual sections instead of repeating them.

1. Install PLECS

1. Download PLECS Standalone (version 5.0.2 or newer) from https://www.plexim.com/download and install it.

2. Get and activate a license.

- Request a student license at https://plexim.com/store/students and follow the directions on the page.
- Use the following course code, 9KH-TrnPsPgN, and register with your @ucalgary.ca email address (other addresses will not work).

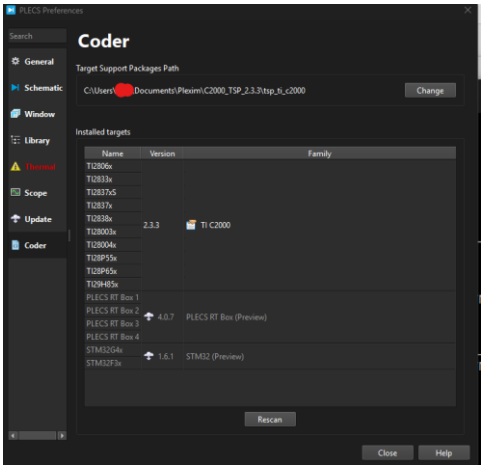
2. Install the TI C2000 Target Support Package

The Target Support Package (TSP) is what teaches PLECS to generate and flash code for the F28P55x. Use TSP 2.3 or newer (2.3.3 is current); TSP 2.3 requires PLECS 5.0.2 or newer.

1. Download the TSP Windows Installer Executable from https://www.plexim.com/download/tsp_c2000.

2. Run it and choose Full Installation. This also downloads the TI tools that PLECS uses behind the scenes: the C2000 compiler (C2000-CGT) and UniFlash, which programs the chip over USB.

3. Open PLECS and go to File > Preferences > Coder. Click the download button next to TI C2000 and install the latest “TI C2000 TSP 2.3.x (Win)” (skip this if step 2 already registered it). When done, it should look like this:



If anything goes wrong, follow Plexim's installation instructions: https://www.plexim.com/support/installation_help/tic2000. The full TI C2000 TSP User Manual is available in PLECS under Help > PLECS Documentation > TI C2000 Target Support.

3. Check the LaunchPad switches and jumpers

The LaunchPad is used in two set-ups in this course: **HIL testing** (on the Typhoon HIL TI LaunchPad interface card, including Lab 0) and **experimental testing** (on the interface board driving the inverter box). **In both, the LaunchPad is powered by the board it is plugged into, not by USB: jumper JP1 stays open and the USB cable carries only flashing and External Mode data.** The switches and jumpers are identical in both set-ups; what differs is how you power the board it sits on.

Switches and jumpers (both set-ups)

These are the positions used in this course for the LAUNCHXL-F28P55X. All except JP1 are Plexim's recommended positions, and all except JP1 and S6 are TI's factory defaults (TI ships JP1 closed and S6 at 0, right). If a board has been fiddled with, put it back to this state:

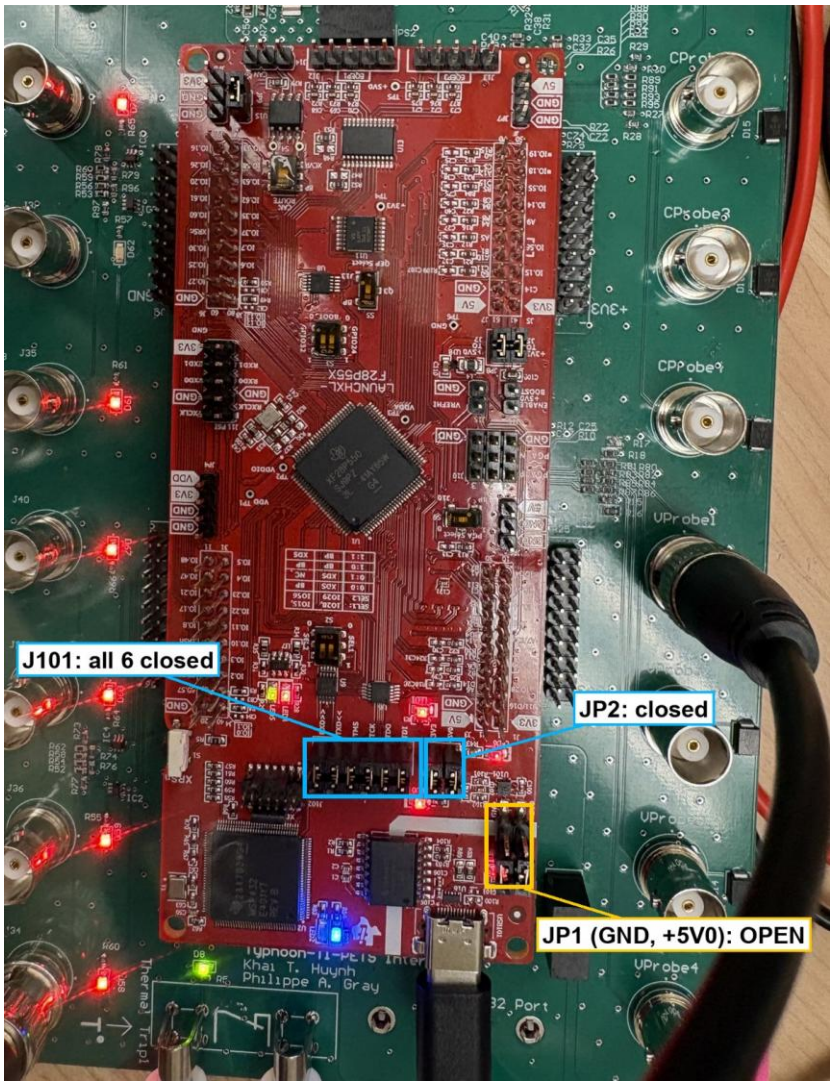
Switch / jumper	Position	What it does
S2 (SEL1, SEL2)	0, 0	Routes GPIO 28/29 to the USB debug probe's COM port - this is the channel External Mode uses
S3 (GPIO24, GPIO32)	1, 1	Boot from flash - the flashed program starts by itself at power-up
S4	Up	Routes GPIO 4/5 to the header (needed for two of the Inverter 2 gate outputs)
S5	Up	Routes GPIO 25/26/30 to the header

S6	1 (left)	Routes the PGA pins to the header
J101 (all six jumpers)	Closed	Connects the serial and JTAG lines to the on-board debugger - required for flashing and External Mode
JP1 (GND, +5V0, next to the USB-C connector)	**Open** — shunts parked on one pin each	Isolates the USB 5 V from the board, so the board is powered from the HIL interface card or the inverter-box interface board. Not the factory default (TI ships it closed).
JP2 (+3V3, +5V0, beside J101)	Closed	Connects the debugger's 3.3 V and 5 V rails to the MCU side of the board

Powering the LaunchPad

	HIL testing (HIL TI LaunchPad interface)	Experimental testing (inverter box)
LaunchPad is powered from	The Typhoon HIL, through the interface card: both HIL POWER switches (5V and 3V3) at ON	The inverter-box interface board, fed from the +15 V auxiliary supply (E3631A)
Before building/flashing or connecting External Mode	Turn on the HIL 101, set both HIL POWER switches to ON and check that LEDs D1 and D2 beside them are lit, then plug in the USB cable	Turn on the +15 V supply and check that interface-board LEDs D5, D9 and D4 are lit, then plug in the USB cable
When power is removed	Turning off the HIL or its HIL POWER switches resets the LaunchPad	Turning off the +15 V supply resets the LaunchPad

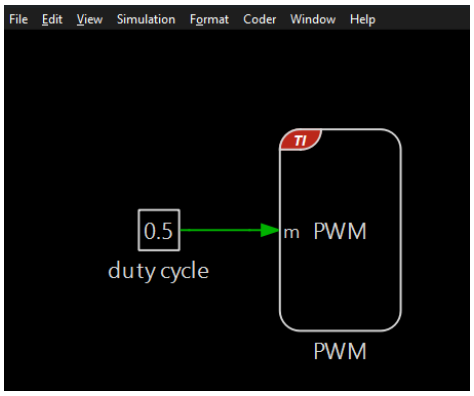
With JP1 open, the USB cable does not power the board: if the interface board is unpowered, flashing and External Mode fail even though the cable is plugged in. After any reset, reconnect External Mode and check your tunable parameters (for example the enable switch) before continuing. **Do not close JP1 while the LaunchPad is on a powered interface board** — the USB 5 V and the interface board's 5 V rail would then be tied together (Typhoon's manual likewise requires all USB power on the LaunchPad to be disabled before powering it from the HIL).



LaunchPad in the experimental set-up, seated on the interface board with +15 V applied: J101 and JP2 closed; JP1 open, with its shunts parked on one pin each. The same jumper settings are used for HIL testing.

4. Generate code and flash the LaunchPad

PLECS does the whole chain from one button: generate C code from your subsystem, compile it, and flash it through the LaunchPad's USB debug probe.



1. Open Coder > Coder options... > Target. Select TI28P55x as the target, set Board to the LaunchPad, Build type to “Build and program”, and Build configuration to “Run from Flash”.
2. Configure the PWM, Protections and ADC settings as needed.
3. Power the board the LaunchPad sits on first (see section 3). Then connect the LaunchPad over USB (data cable) and click Build. PLECS generates the code, compiles it, flashes the MCU, and the program starts running immediately.

If programming fails, open the PLECS diagnostics window (the exclamation icon at the bottom-right of the schematic window) for the error details.

Note: the lab files contain only the controller, so PLECS builds the whole model as-is. If your own model (for example for the final project) also contains a simulated plant, put the controller and the peripheral blocks inside a subsystem and enable code generation for it (right-click the subsystem > Subsystem > Execution settings... > enable code generation) - that subsystem is what gets flashed, while the rest of the model stays simulation-only. This lets you simulate the closed loop and deploy the same controller without maintaining two files.

5. External Mode: live scopes and parameter tuning

External Mode connects PLECS to the running MCU over the same USB cable: Scopes in your model display real-time data from the target, and you can change tunable parameter values (for example controller gains or references) while the code runs - no re-flash needed. Changing the circuit or model structure still requires a rebuild.

1. Before building, open Coder options... > Target > External Mode and check the interface is set to Serial with GPIO 28 (Rx) / GPIO 29 (Tx) - the default for this LaunchPad.

2. With the code running, go to Coder options... > External Mode, click the edit icon next to the Target device field, then Scan, and pick the LaunchPad's serial port. The XDS110 debug probe shows up as two COM ports - use the one named “XDS110 Class Application/User UART”, not the “Auxiliary Data Port”.

3. Click Connect. Set Number of samples (e.g. 8192) and click Start autotriggering - your Scopes now show live MCU data. Set a Trigger channel to synchronize the capture to an event, just like an oscilloscope.

4. Change any tunable parameter in the model dialogs and it takes effect on the target immediately. Only parameters listed as inlining exceptions (Coder options... > Parameter Inlining) can be changed; by default PLECS inlines every other parameter value into the code. The lab files already list theirs (for example the enable switch and the modulation command). In your own model, add the gains and references you want to tune to that list before you build. The model is locked against structural edits such as rewiring while connected; click Disconnect when done.

6. Resources

Resource	Where
PLECS download	https://www.plexim.com/download
TI C2000 Target Support Package download	https://www.plexim.com/download/tsp_c2000
TSP installation instructions	https://www.plexim.com/support/installation_help/tic2000
TI C2000 TSP User Manual	In PLECS: Help > PLECS Documentation > TI C2000 Target Support (also on the Plexim website)
Demo models	Installed with the TSP; open via File > Demo Models in PLECS
LaunchPad hardware details	TI LAUNCHXL-F28P55X User's Guide (SPRUJC0)